THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Assertion (A): A fair test changes only one factor at a time. Reason (R): Changing several factors together makes it impossible to know which caused the effect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Animal cells burst more easily than plant cells in very dilute surroundings. Reason (R): Animal cells have a rigid cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Antibiotics are not effective against influenza. Reason (R): Influenza is caused by a virus, which has no bacterial structures for antibiotics to act on. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): An electromagnet loses its magnetism when the current is switched off. Reason (R): Soft iron does not retain magnetism. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): It is easier to push a box on a floor with rollers than on a rough floor. Reason (R): Rolling friction is less than sliding friction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): A knife with a sharp edge cuts easily. Reason (R): A sharp edge exerts large pressure because of its small area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): Hot tea cools faster when poured in a wide saucer. Reason (R): A larger surface area increases the rate of evaporation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): Water is a compound and not a mixture of hydrogen and oxygen. Reason (R): Hydrogen and oxygen in water are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): A solution of sugar in water does not scatter a beam of light. Reason (R): The particles of the solute are of very small size. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The image formed by a plane mirror is virtual. Reason (R): It cannot be obtained on a screen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): The shadow of a vertical stick is shortest at noon. Reason (R): At noon the Sun is highest in the sky. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Removing top carnivores from a forest can harm the whole ecosystem. Reason (R): Herbivore numbers increase and overgraze the vegetation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): The Earth's inner core is solid even though it is extremely hot. Reason (R): The very high pressure at the centre prevents it from melting. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. A student studies the cooling of tea. Temperatures at 0, 2, 4, 6 minutes are 90, 78, 69, 62 degrees C. The fall during the first 2 minutes compared with the last 2 minutes is: 1

(a) equal **(b)** smaller **(c)** zero **(d)** greater

15. A bacterium of length 2 micrometres is viewed with total magnification of 1000x. Its image size is: 1

(a) 20 mm **(b)** 2 mm **(c)** 2000 mm **(d)** 0.2 mm

16. In a school of 500 students, 40 got infected with influenza in a week and 30 more next week. The number infected after two weeks is: 1

(a) 30 **(b)** 40 **(c)** 70 **(d)** 500

17. Two nichrome wires A and B are of equal length; A is thicker. The same current flows in both. Which statement is correct? 1

(a) A gets hotter than B **(b)** B gets hotter than A
(c) both stay cold **(d)** A melts first

18. A box of mass 20 kg is pulled with a force of 60 N along a floor where friction is 40 N. The net force is: 1

(a) 40 N **(b)** 100 N **(c)** 20 N **(d)** 60 N

19. A cylinder of weight 120 N stands on a base of radius 10 cm. Taking pi as 3.14, the pressure is about: 1

(a) 38220 Pa **(b)** 12 Pa **(c)** 1200 Pa **(d)** 3822 Pa

20. 100 mL of water and 100 mL of ethanol are mixed. The volume of the mixture is slightly less than 200 mL. This best illustrates: 1

(a) condensation **(b)** evaporation
(c) spaces between particles **(d)** sublimation

21. 4 g of hydrogen combines with 32 g of oxygen to form water. The mass of water formed and the ratio of hydrogen to oxygen are: 1

(a) 36 g, 1 : 4 **(b)** 28 g, 1 : 8 **(c)** 36 g, 1 : 8 **(d)** 36 g, 8 : 1

22. At 40 degrees Celsius, 100 g of water dissolves 64 g of potassium nitrate in a saturated solution. The mass percentage of the saturated solution is about: 1

(a) 64 per cent **(b)** 36 per cent **(c)** 100 per cent **(d)** 39 per cent

23. The angle between a ray and a plane mirror is 35 degrees. If the mirror is rotated by 15 degrees so that the angle of incidence increases, the new angle between the incident and the reflected ray is: 1

(a) 110 degrees **(b)** 100 degrees **(c)** 140 degrees **(d)** 70 degrees

24. A pole of height 3 m has a shadow of 4 m and a nearby tree has a shadow of 20 m at the same time. The height of the tree is: 1

(a) 15 m **(b)** 26 m **(c)** 10 m **(d)** 20 m

25. In the food chain grass, grasshopper, frog, snake, the energy available is least at the level of the: 1

(a) frog **(b)** grass **(c)** grasshopper **(d)** snake

26. The core of the Earth has a radius about 3400 km and the Earth about 6400 km. The ratio of the core radius to the Earth radius is about: 1

(a) 0.64 **(b)** 1.88 **(c)** 0.34 **(d)** 0.53

27. A thermometer reads 3 degrees C in melting ice. A reading of 53 degrees C for warm water corresponds to a true value of: 1

(a) 53 degrees C **(b)** 47 degrees C **(c)** 50 degrees C **(d)** 56 degrees C

28. A cell loses its nucleus. The most probable result is that the cell: 1

(a) forms chloroplasts **(b)** grows faster

(c) becomes a plant cell **(d)** stops functioning properly

29. A student claims, 'I took antibiotics for my cold and it got better, so antibiotics cure colds.' The flaw is: 1

(a) colds are viral and often recover by themselves

(b) antibiotics are vaccines

(c) colds are bacterial

(d) antibiotics cure all diseases

30. A coil of wire is wound on a plastic tube and then on an iron rod, with the same current. The magnetic effect is stronger with: 1

(a) equal **(b)** the iron rod

(c) the plastic tube **(d)** neither, as there is no magnetic effect

31. A tug-of-war rope is stationary. If team A suddenly pulls harder, then the rope: 1

(a) stays at rest **(b)** moves towards team B

(c) moves towards team A **(d)** breaks always

32. Two cyclones, A over warm sea and B moving over land, are compared. Which statement is right? 1

(a) B is likely to be stronger **(b)** A is likely to be stronger

(c) both are equal **(d)** neither has wind

33. A student observes that a drop of food colour spreads equally in a glass of still water but faster in warm water. The explanation is: 1

(a) temperature increases the motion of particles

(b) gravity acts differently

(c) colour is lighter

(d) water is thicker

34. A coin is made of an alloy of copper and nickel. This coin is a: 1

(a) element **(b)** homogeneous mixture

(c) heterogeneous mixture **(d)** compound

35. A liquid X mixes with water in all proportions while liquid Y does not. Which can be separated from water with a separating funnel? 1

(a) X **(b)** Y **(c)** both **(d)** neither

36. Light incident on a smooth metal surface at 90 degrees to the surface is reflected at: 1

(a) 90 degrees to the surface (back along normal)

(b) 180 degrees

(c) 45 degrees

(d) 0 degrees

37. The Moon appears the same size as the Sun in the sky although the Sun is about 400 times larger because the Sun is also: 1

(a) closer **(b)** about 400 times farther

(c) dimmer **(d)** smaller

38. Which of the following is a good example of mutualism? 1

(a) tiger and deer **(b)** mosquito and human

(c) tapeworm and human **(d)** bees and flowers

39. A glacier's melting adds water to the oceans. If all land ice melted, the effect on coastal cities would be: 1

(a) no change **(b)** flooding by rising sea levels

(c) cooler climate **(d)** drought on the coast

40. Assertion (A): An experiment should be repeated several times. Reason (R): Repetition ensures that the hypothesis is always true. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. **Assertion (A):** In a fair test on how oil temperature affects the puffing of a puri, the thickness of the dough must be kept the same. 1
Reason (R): Only one variable should be changed at a time while all other factors are kept constant.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Robert Hooke used the name 'cells' for the box-like compartments he saw in a thin slice of cork. 1
Reason (R): Cork is made of dead cells whose walls remain as empty compartments.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Health means only the absence of disease. 1
Reason (R): Health includes physical, mental and social well-being.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** A compass needle placed near a wire deflects when current flows through the wire. 1
Reason (R): An electric current produces a magnetic field around the wire.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Friction is a contact force. 1
Reason (R): Friction acts only when two surfaces are in contact and opposes their relative motion.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** A sharp knife cuts better than a blunt one. 1
Reason (R): For the same force, a smaller area of contact gives a larger pressure.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Sugar dissolves in water without much increase in the volume of water. 1
Reason (R): There are spaces between the particles of water into which the sugar particles fit.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Oxygen is an element. 1
Reason (R): Oxygen contains only one kind of atom and cannot be broken down into simpler substances by chemical methods.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A solution of salt in water does not scatter a beam of light. 1
Reason (R): The particles in a true solution are extremely small.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The image formed by a plane mirror cannot be obtained on a screen. 1
Reason (R): The image in a plane mirror is virtual.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**

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